Bryce Hoenigman

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EDUCATION

Johns Hopkins Center for Talented Youth, *Baltimore, MD*
Cumulative GPA: 4.0

- Completed 5 year Modern Standard Arabic program

San Francisco University High School, *San Francisco, CA*
Cumulative GPA: 3.96

- Linguistics coursework: Faculty sponsored independent studies in the Foundations of Linguistics and Historical Linguistics, Wild Tongues; Sociolinguistics and the Philosophy of Language
- Other relevant coursework: AP Statistics (5)

The University of Chicago, *Chicago, IL*
Cumulative GPA: 3.88

- Linguistics coursework: Introduction to Linguistics, Semantics and Pragmatics, Phonetics and Phonology, Morphology, Computational Linguistics I-II, Mind Brain Meaning, Language and the Human I-III, Old Babylonian I-III, Comparative Semitics
- CS coursework: Introduction to Python, Introduction to Data Science I; Computation and Statistics, Introduction to Data Science II; Analysis and Machine Learning, Data Visualization, Computational Linguistics, AI Development Program, Machine Learning: Concepts and Applications.

EXPERIENCE

Volunteer, NCL Board Role Appointee
National Charity League, Golden Gate Chapter • 2017 - 2024

- Board role appointee (VP of Communications 2022-2023, President 2021-2022, Parliamentarian 2019-2021).
- Over 50 hours of service each year with NCL–built experience serving the bay area community through over 30 local organizations.
- Recipient of the Silver-Covid and Bronze Presidential Service Award.

VP of Academics
University High School, San Francisco • 2022 - 2023

- Organized school-wide book drives to make necessary resources accessible to students.
- Worked closely with the Dean of Academics and Dean of Student Life to bring ideas for improving academic life at the school into fruition.
- Launched and managed the peer tutoring canvas page, creating resources within it and overseeing discussion board post questions.

Lead Peer Tutor
University High School, San Francisco • 2020 - 2024

- Tutored students in English, Algebra, Calculus i/ii/iii, Statistics, Organization/Study Skills.
- Proctored study halls for drop-in tutoring.
- TA-ed in Western Civilization (history/art/music) and led weekly study sessions.
- Created spaces to review foundational math that students missed due to COVID-19.

Quad Faculty Research Grant; Tablet Collection
Institute for the Study of Ancient Cultures, Chicago • 2025-

- Identifying and photographing tablets to aid researchers.
- Photoshopping tablets into fat crosses to upload to ISAC's database.
- Efficiently processing and corresponding to external photo requests on sheets and via email.
- Conducting research through photography, object analysis, and supplementary texts on prisms during the reign of Ashurbanipal.
- Aiding in the Tablet Collection's Nippur publication project by transliterating, normalizing, translating, and creating hand copies for my tablet assignments.

Script Encoding Initiative, Summer Intern
University of California Berkeley • 2025

- Compiled list of relevant sources on Zero-Width-Joiners (ZWJs) and Zero-Width-Non-Joiners (ZWNJs) in Zotero.
- Requested boxes to pull from the Stanford Special Collections in Silicon Valley archive, spent two days in person sorting, scanning, and uploading all relevant documents.
- Synthesized research into an informative document on the role of ZWJs and ZWNJs across bidirectional scripts, Indic scripts, and emoji.
- Wrote and published an article for SEI's blog on my research and its importance for script encoding.

Emergent Ventures Grantee, Archaeology Tranche
Mercatus Center at George Mason University • 2026

- Built an image classification pipeline that predicts where a cuneiform tablet was written from 158,946 sign images
- Fine-tuned ResNet101 and EfficientNetV2-S encoders on 266-class sign identification, reaching 84.3% accuracy and 0.764 macro-F1
- Trained EfficientNet-B0 provenance classifiers: 0.95 macro-F1 for north vs. south and 0.68 macro-F1 across 16 cities (baseline 0.017)
- Won the computing award at MindByte's RCC research symposium

Digital Pasts Summer Research Intern
Digital Pasts Laboratory • 2026

- Analyzed Akkadian-determinative disambiguation metrics by fine-tuning a roBERT masked learning model and seq2seq translation model
- Reduced data migration friction by designing a SQL database schema to migrate data from ORACC into the iClassifier project
- Extracted determinatives from ORACC Akkadian texts in Python and ran statistical analysis to identify corpus-wide usage patterns
- Reviewed prior literature to select candidate variables for statistical modeling of determinative usage
- Serving on ALP2027 Student Board

RELEVANT ACTIVITIES

UChicago Linguistics Undergraduate Board (CLUB), Chicago, IL

- ❖ Board of five students who organize linguistics-related speaker events and publications.

Conlang Club Minister, Chicago, IL

- ❖ A group of students who participate in constructing new languages using linguistic knowledge.

AI@UChicago, Chicago, IL

- ❖ Student-led classes in learning to code basic Machine Learning concepts (decision trees, clustering algorithms, neural networks, and backpropagation) and apply them through hands-on AI-centered projects.
    - ➢ Zero-to-Transformer Project (*Winter, 2026*): project starting from foundational knowledge in machine learning and ending in building a transformer from scratch. This project will result in being able to build a full transformer model through exploring concepts like self-attention, encoder-decoder architectures, and modern transformer variations.

Nahuatl, Chicago, IL

- ❖ Student-led courses in the Uto-Aztecan language of the Nahuatl.

SKILLS

Python (libraries: sci-kit-learn, matplotlib, pandas, stats models, seaborn, numpy, altair, pytorch, pydantic)
SQL (flavor: sqlite, schema design/DDL)

LANGUAGES

Limited working knowledge of Arabic
Limited working knowledge of Spanish
Limited working knowledge of Akkadian
Elementary Nahuatl
Elementary Italian